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Neural Network based Emulation of Subgrid-Scale Turbulence in MONC: Bridging Offline Multi-Task Learning and Online Coupling with FTorch

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1 Motivation & Background

- ❑ LES resolves the highest energy-containing eddies with sub-filter mixing & dissipation needing an SGS closure.
- ❑ In MONC, the **Smagorinsky–Lilly** scheme handles the SGS closure
- ❑ Goal: To develop a **robust, accurate, stable, efficient and resolution-invariant** neural-network (NN) emulator of this closure, coupled online inside MONC.
- ❑ Approach: Use **Multi-Task Learning (MTL) approach** for joint representation learning

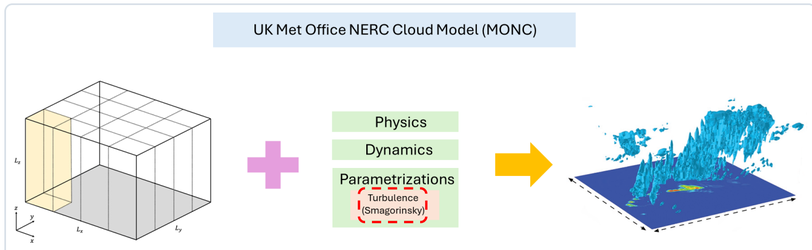


Fig. 1: Schematic representation of MONC LES set-up: dynamics + physics + parameterizations (incl. Smagorinsky turbulence).

Richardson Number (Ri): Represents turbulence intensity and stability

$$\mathrm{Ri} = \frac{(g/\theta_0)(\partial\bar{\theta}/\partial z)}{\left(\frac{\partial\bar{u}}{\partial z}\right)^2 + \left(\frac{\partial\bar{v}}{\partial z}\right)^2}$$

Ri discriminates the turbulence regime

Ri < 0 : unstable — buoyancy-driven turbulence

0 < Ri < 0.25 : stable — mechanical mixing

Ri ≥ 0.25 : turbulence suppressed entirely

MONC SGS scheme computes eddy viscosity K_m & diffusivity K_h, modulated by stability functions f(Ri)

2 Research Questions

- ❖ **RQ1:** Is multi-resolution training from limited sparse, stencil-based points/samples feasible, with full-3D inference capability?
- ❖ **RQ2:** Do physics-guided frameworks beat data-driven ones at equal training data?
- ❖ **RQ3:** Should physics persist at inference (explicit - through conditioning) or act only during training (implicit - through gradients)?
- ❖ **RQ4:** Is MTL the right fit and do CV task-weighting methods transfer to complex coupled physics simulations?
- ❖ **RQ5:** Does architectural complexity help or hinder cross-regime generalization, under limited data regime?

3 Data & Methodology

- ➢ **Two atmospheric regimes, 20× resolution gap:** Radiative Convective Equilibrium (RCE) deep convection (Δx = Δy = 1 km) + Atmopsheric Radiation Measurement (ARM) shallow cumulus (Δx = Δy = 50 m).
- ➢ **Feature Engineering:** 54 input features per point: star-stencil neighbourhood, prognostic variables, grid metadata & geometry.
- ➢ **Regime-stratified sampling:** 4.05 M training points (~5% of available data; ~24% RCE / ~2% ARM domain) vs 202.8 M evaluation points.
- ➢ **Multi-task learning:** shared backbone (MLP / ResMLP / TabTransformer) + heads for K_m, K_h, Ri and regime — 54 configs (6 strategies × 3 architectures × 3 seeds).

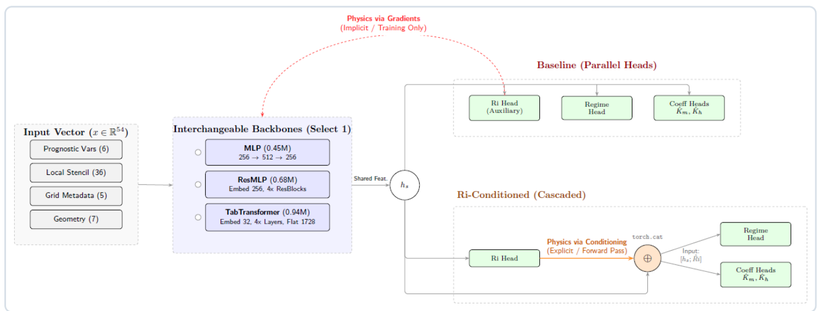


Fig. 2 MTL set-up: interchangeable backbones; physics via gradients (auxiliary tasks) or via forward-pass Ri-conditioning.

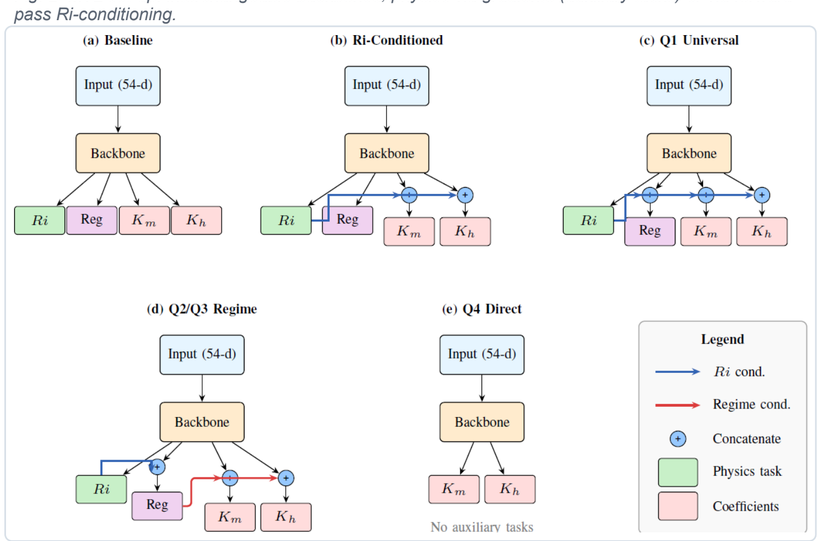


Fig. 3 Six physics-integration strategies: Baseline (gradients only), Ri-Conditioned, Q1 universal conditioning, Q2/Q3 cascades, Q4 data-driven ablation.

The two regimes at a glance

RCE
Deep tropical convection · Δx = Δxy= 1 km · 66% of training samples — the data rich regime.

ARM
Shallow mid-latitude cumulus · Δx = Δy = 50 m · 34% of training samples— the distribution-shift testbed (data poor regime).

Note: *the two configurations use different Smagorinsky stability-function constants — the same Ri maps to different coefficients. Ri prediction is universal; coefficient prediction is configuration-dependent.*

4 Task Weighting: Uncertainty method Wins

Weighting	MLP	ResMLP	TabTransformer
Manual	0.626 ± .031	0.708 ± .004	0.633 ± .007
Uncertainty	**0.835 ± .004**	**0.869 ± .005**	**0.821 ± .004**
DWA	0.601 ± .047	0.300 ± .013	0.258 ± .014

Table 1 Test-split viscosity R^2 scores for 3D inference (mean ± std over 3 seeds) using Baseline configuration.

- ❖ **Uncertainty weighting wins in all 54 configurations.**
- ❖ DWA is oscillatory across every seed and collapses badly on deeper networks — not all CV weighting methods can be applicable to turbulence SGS emulation

5 Cross-Regime Generalization

Model (R^2)	RCE Visc	RCE Diff	ARM Visc	ARM Diff
Q3-MLP	**0.811**	**0.808**	0.616	0.701
Baseline-MLP	0.801	0.798	0.540	0.644
Q4-MLP (data-driven)	0.784	0.782	0.642	0.645
Ri-MLP	0.773	0.770	**0.670**	**0.734**
Q1-MLP (universal)	0.722	0.721	**0.119**	0.424

Table 2 Full-domain R^2, MLP variants. RCE = dominant / in-distribution; ARM = rare / out-of-distribution.

- Data-rich (RCE): baseline wins. Data-poor (ARM): Ri-MLP wins — **+24% over Baseline;** explicit conditioning helps precisely under distribution shift.
- **Q1 collapses (R^2 0.119):** universal conditioning cascades Ri errors into the coefficient heads — targeted physics conditioning beats naive expansion.
- Ri-conditioning yields **+158% active-turbulence viscosity R^2 scores (ARM)** over the purely data-driven end member (Q4).

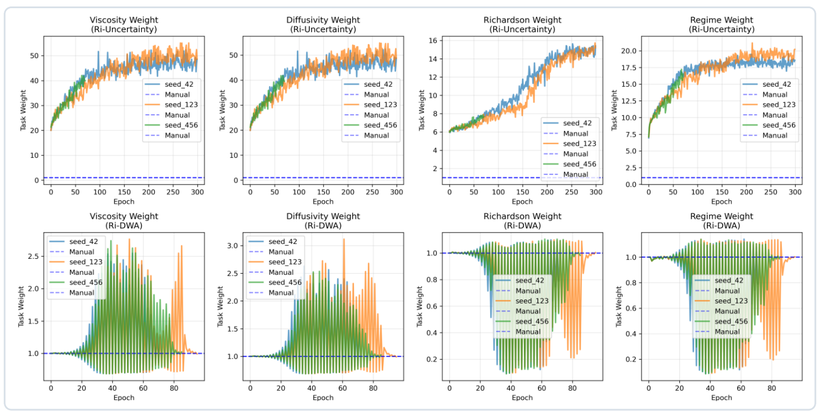


Fig. 4 Task-weight evolution during MTL training (Ri-MLP, 3 seeds): uncertainty weighting (top row) converges smoothly for all four task weights; DWA (bottom row) oscillates persistently — consistent with uncertainty weighting winning in all 54 configurations.

6 Error Anatomy (KGE Decomposition)

Data · Model	KGE	r (Corr. Coeff)	α (var. ratio)	β (bias ratio)
RCE · Q3-MLP	0.809	0.901	0.905	1.13
ARM · Ri-MLP	0.426	0.845	0.885	**1.54**
ARM · Baseline-ResMLP	−0.025	0.487	**1.80**	1.39

$$KGE = 1 - \sqrt{[(r-1)^2 + (\alpha-1)^2 + (\beta-1)^2]}$$

- ❑ **MLPs are bias-dominated** (β) → fixable with more / better-balanced data. **ResMLPs are variance-dominated** (α) → need constrained residual pathways.
- ❑ Full 3D inference shows high correlation everywhere → the physics is learnt. Accuracy of MTL models limited by data coverage, which can be improved.

7 Robustness Under Distribution Shift

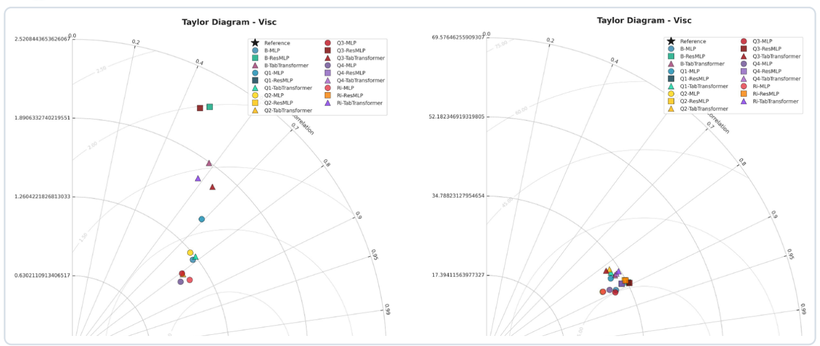


Fig. 5 Taylor diagrams (viscosity): ARM (left), RCE (right). All architectures tie in-distribution; under the ARM shift only MLP variants (circles) keep realistic variance & correlation — ResMLP / TabTransformer show variance explosion (up to 5.3×).

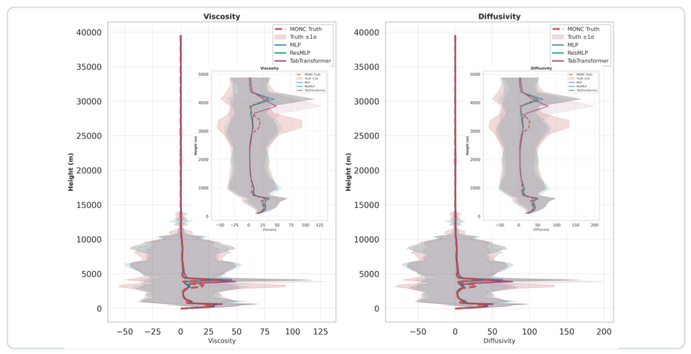
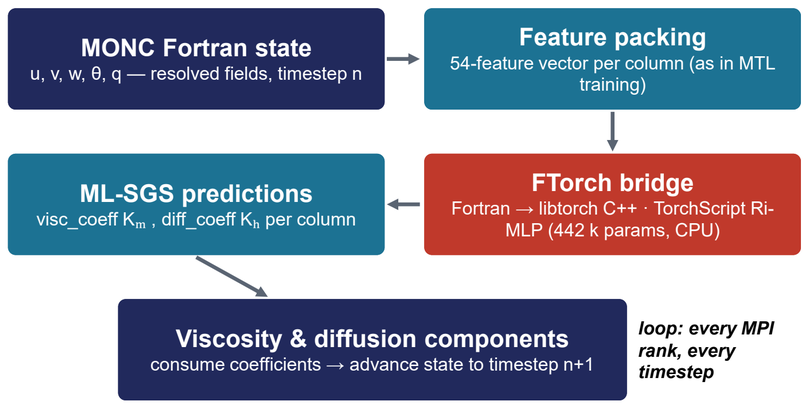


Fig. 6 Offline full-domain reconstruction: domain-averaged vertical profiles from Ri-conditioned models (shading = per-level variability) — one MTL based NN SGS model reconstructs full 3D domain with great accuracy.

8 Online Coupling inside MONC via FTorch



Deployment (RACC2 cluster at University of Reading): Apptainer container (gfortran · FTorch · libtorch CPU · NetCDF/HDF5/MPI) — no Python at runtime. 64 MPI ranks/node; SLURM job-chaining with restart continuation for long runs; TorchScript model + scalers bind-mounted from host.

9 Online Fidelity: ML-SGS vs Smagorinsky

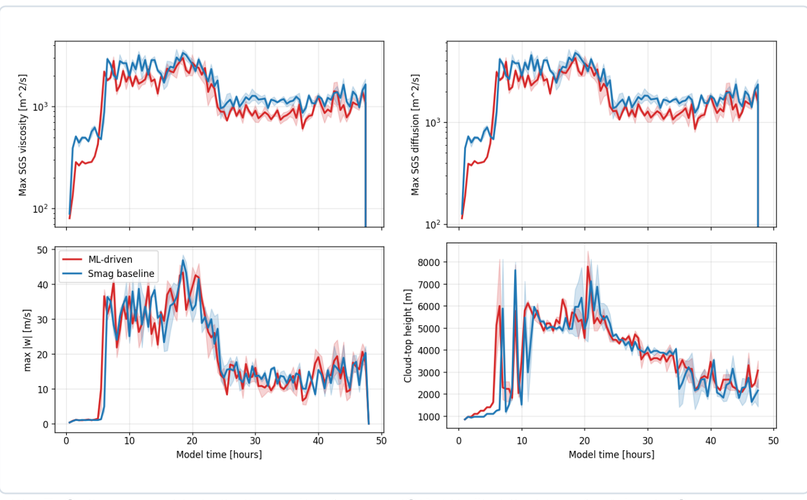


Fig. 7 Online ensemble bulk diagnostics (mean ± IQR, N ≥ 3): ML-driven (red) tracks Smagorinsky (blue) through the full 48 h of RCE simulation online (time axis: 0–48 h).

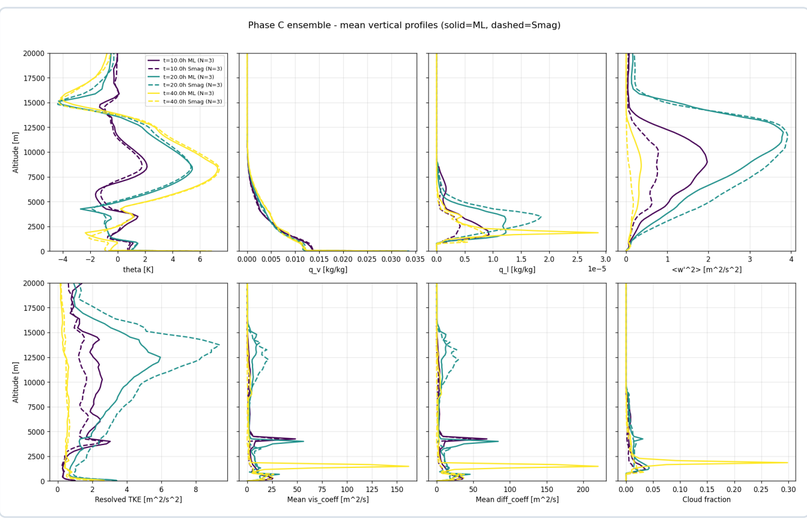


Fig. 8 Ensemble-mean vertical profiles at t = 10 / 20 / 40 h (solid = ML, dashed = Smag): realistic thermodynamics, TKE and SGS coefficients at every snapshot. Stable — no NaN / blow-up over 48 h and beyond (25 days of chained model time; ~450 h of ML-driven simulation analysed).

10 Spectral Fidelity & Cross-Resolution Performance (RCE Regime)

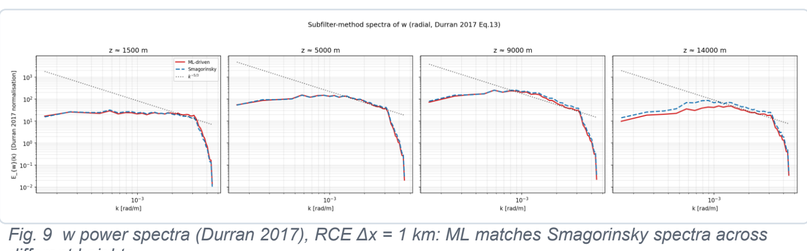


Fig. 9 w power spectra (Durran 2017), RCE Δx = 1 km: ML matches Smagorinsky spectra across different heights.

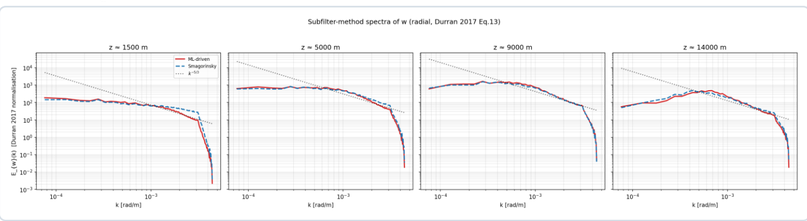


Fig. 10 Unseen resolution, Δx = 500 m (not in training): the ML-SGS-driven MONC still produces realistic spectra while closely emulating Smagorinsky — potential towards of resolution invariance!

11 Conclusions

- ❑ **Initial findings seem promising. MTL based neural emulation of SGS works!**
- ❑ **Stable online simulations: no NaN/blow-up over a full 48 h diurnal cycle (and beyond!). Also confirmed with ensemble simulations.**
- ❑ **Realistic physics: ML-SGS (Ri-MLP based config) tracks Smagorinsky across every bulk diagnostic and at every profile snapshot.**
- ❑ **Offline training of ML-SGS to stable online-coupling is possible, with careful design and physics-guidance.**
- ❑ **Performance benchmarking across various atmospheric regimes is ongoing and could require some system (MONC) level optimization.**
- ❑ **Future Work: Scale the model training with additional data and intercompare all MTL configs online.**

Key Takeaways

1. Multi resolution training from sparse samples → full 3D domain inference works
One model spans a 20× resolution gap from 4 M sparse points — and runs stably online, even at unseen resolutions and timesteps.

2. Targeted physics integration wins (under limited data regimes)
Ri-conditioning: +158% improvement in ARM active-turbulence viscosity R^2 scores over the data-driven end member under equal training data.

3. Physics should persist at inference (conditioning), for managing distribution shifts
Architectural explicit conditioning beats gradients-only approach, but only if applied selectively: universal conditioning collapses (R^2 = 0.12 in ARM cases).

4. MTL works for SGS emulation; not all CV weightings are applicable
Uncertainty weighting wins by 20–30% in all 54 configs; DWA oscillates and collapses on deeper, more complex neural network configurations.

5. Simple configs generalize well with proper physics-guidance, under limited data regime
All MTL configs tie on RCE; only MLP variants survive the ARM shift — complex models show variance explosion.

Models · Data · Code

Zenodo · WCCI

GitHub · WCCI

Zenodo · PASC

GitHub · PASC

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